



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## ENCRYPTION PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Security

TOTAL: 10 QUESTIONS

**Q1** What is Encryption and what security threat does it address?

Threat Model

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**Q6** How do you audit and monitor Encryption events?

Observability

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**Q2** What is the core mechanism Encryption uses to provide security?

Architecture

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**Q7** How does Encryption handle key or credential rotation?

Lifecycle

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**Q3** How do you configure Encryption for a production web application?

Configuration

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**Q8** How does Encryption help meet compliance requirements?

Compliance

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**Q4** What are the most common Encryption misconfigurations to avoid?

Hardening

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**Q9** How do you test your Encryption configuration for weaknesses?

Testing

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**Q5** How does Encryption integrate with identity providers?

SSO / IdP

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**Q10** What are the performance implications of Encryption and how do you minimize them?

Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Encryption is a security tool, protocol,

Mitigation

Encryption is a security tool, protocol, or service that mitigates a specific class of threats authentication bypass, data exfiltration, network intrusion, or credential theft. Understanding the threat model is prerequisite to correct configuration.

A6 Enable Encryption's audit log and stream

Observability

Enable Encryption's audit log and stream it to a SIEM (Splunk, Elastic SIEM). Define alert rules for high-severity events: repeated failed logins, privilege escalation, and unusual access patterns. Review logs regularly.

A2 Encryption uses one or more security

Primitives

Encryption uses one or more security primitives: asymmetric cryptography, symmetric encryption, certificate chains, role-based access control, or anomaly detection. These primitives are combined to provide the claimed security properties.

A7 Automate rotation using a secrets manager

Automation

Automate rotation using a secrets manager that generates new credentials and updates consumers transparently. Test rotation in staging. Have a break-glass procedure for emergency rotation when a credential is compromised.

A3 Production configuration requires strong cipher

Hardening

Production configuration requires strong cipher suites, certificate rotation, revocation checking, and minimal permission grants. Use a well-reviewed configuration reference (CIS Benchmark, OWASP) rather than default settings.

A8 Encryption generates audit trails, enforces

Governance

Encryption generates audit trails, enforces access policies, and provides encryption that satisfy controls required by SOC 2, PCI-DSS, HIPAA, and GDPR. Map Encryption's capabilities to the specific framework controls in your compliance program.

A4 Common mistakes include using outdated

Gotchas

Common mistakes include using outdated algorithms (MD5, SHA1, RC4), leaving default credentials, ignoring certificate expiry, granting excessive permissions, and not testing the config before going live in production.

A9 Run an automated scanner (SSL Labs

Assessment

Run an automated scanner (SSL Labs for TLS, ScoutSuite for cloud IAM, OWASP ZAP for web app auth) against your Encryption config. Conduct penetration tests annually and after significant config changes.

A5 Encryption delegates identity verification to an

Federation

Encryption delegates identity verification to an IdP via SAML, OIDC, or LDAP. Users authenticate once (SSO) and receive a token that Encryption validates. This centralizes user management and avoids duplicating auth logic across services.

A10 Cryptographic operations, token validation, and

Optimization

Cryptographic operations, token validation, and authorization checks add latency. Mitigate with session caching, hardware-accelerated TLS (AES-NI), connection reuse, and async authorization where possible.